

1.22 Project - RDBMS

3.15 Project Deliverable 1

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Project - RDBMS

1. Identify the system that you want to implement.

A personal financial application. Think Quicken, but customized more specifically for a monthly profit & loss statement.

2. Analyze all the system's requirements.

It should track cash, credit card, and bank transactions. It should be able to produce reports on transactions by categories, accounts, banking companies, dates, etc. It should use procedural logic, data patterns, and predictive modeling to assign categories to transactions with limited user input.

3. Identify the list of entities and attributes.

- Transactions
 - Transaction date
 - Amount
 - Merchant
 - Account
 - Person who transacted
- Accounts
 - Account name
 - Account number
 - Account credentials
 - Account type
- Institutions
 - Institution name
- Rules
 - Rule type
 - Business logic
- Categories
 - Category top level
 - Category mid level
 - Category sub level
- Merchants
 - Merchant name
 - Merchant category

4. Perform normalization process.

In #3 we were asked to provide entities and attributes; when we list these, we are automatically doing this in a third normal form. If #3 was really asking for a brainstormed list of data we would list the following:

Transactions

Credit Card Number, Date, Amount, Merchant

Then we would normalize by adding Account, Institution, Categories and Merchant entities.

5. Identify the few PK/FK.

It is best practice to have a monotonically increasing integer **Primary Key** on each table with the suffix _id:

Table	Primary Key
Transaction	transaction_id
Account	account_id
Merchant	merchant_id
Category	category_id
Institution	institution_id
Rule	rule_id

Foreign Keys:

Table	Foreign Key Reference Table	Foreign Key
Transaction	Account	account_id
Transaction	Category	category_id
Transaction	Merchant	merchant_id
Transaction	Rule	rule_id
Account	Institution	institution_id

6. Identify the relationships between entities.

Table	Relationship Type	Related Table
Account	one to many	Transaction
Category	one to many	Transaction
Merchant	one to many	Transaction
Rule	one to many	Transaction
Institution	one to many	Account

7. Create the draft ERD diagram

